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Task:

1. Open Git bash shell and create a new folder “**GitDemo**” and initialize as git repository

Solution:

MINGW64:/d/gittraining/GitDemo

```
narendrakumar.s@PTPLL614 MINGW64 /d/gittraining (main)
$ mkdir GitDemo
```

```
narendrakumar.s@PTPLL614 MINGW64 /d/gittraining (main)
$ cd GitDemo

narendrakumar.s@PTPLL614 MINGW64 /d/gittraining/GitDemo (main)
$ git init
Initialized empty Git repository in D:/gittraining/GitDemo/.git/
```

Output:

Clipboard					
Organise					
New					
Open					
Select					
↑ > This PC > New Volume (D:) > gittraining >					
	Name	Date modified	Type	Size	
access					
top	.git	21-07-2025 04:34 PM	File folder		
downloads	GitDemo	21-07-2025 05:07 PM	File folder		
images	sample 2	21-07-2025 11:01 AM	Text Document	1 KB	
res	sample	21-07-2025 10:33 AM	Text Document	1 KB	
	test	21-07-2025 10:33 AM	Text Document	1 KB	

2. Git bash initializes the “**GitDemo**” repository

```
narendrakumar.s@PTPLL614 MINGW64 /d/gittraining/GitDemo (main)
$ git init
Initialized empty Git repository in D:/gittraining/GitDemo/.git/
```

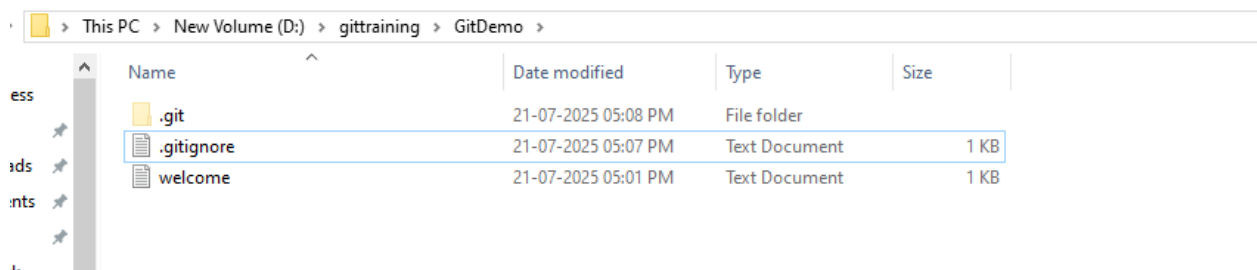
3. Create a file “**welcome.txt**” and add content to the file (use echo or cat command)

Solution:

```
narendrakumar.s@PTPLL614 MINGW64 /d/gittraining/GitDemo (master)
$ git branch -M main

narendrakumar.s@PTPLL614 MINGW64 /d/gittraining/GitDemo (main)
$ echo "Welcome to Git Demo">welcome.txt
```

Output::



Name	Date modified	Type	Size
.git	21-07-2025 05:08 PM	File folder	
.gitignore	21-07-2025 05:07 PM	Text Document	1 KB
welcome	21-07-2025 05:01 PM	Text Document	1 KB

4. To verify if the file “welcome.txt” is created, execute ls-al

Solution:

```
narendrakumar.s@PTPLL614 MINGW64 /d/gittraining/GitDemo (main)
$ ls -al
total 10
drwxr-xr-x 1 narendrakumar.s 1049089  0 Jul 21 17:22 ./
drwxr-xr-x 1 narendrakumar.s 1049089  0 Jul 21 16:53 ../
drwxr-xr-x 1 narendrakumar.s 1049089  0 Jul 21 17:08 .git/
-rw-r--r-- 1 narendrakumar.s 1049089   6 Jul 21 17:07 .gitignore
-rw-r--r-- 1 narendrakumar.s 1049089  47 Jul 21 17:01 welcome.txt
```

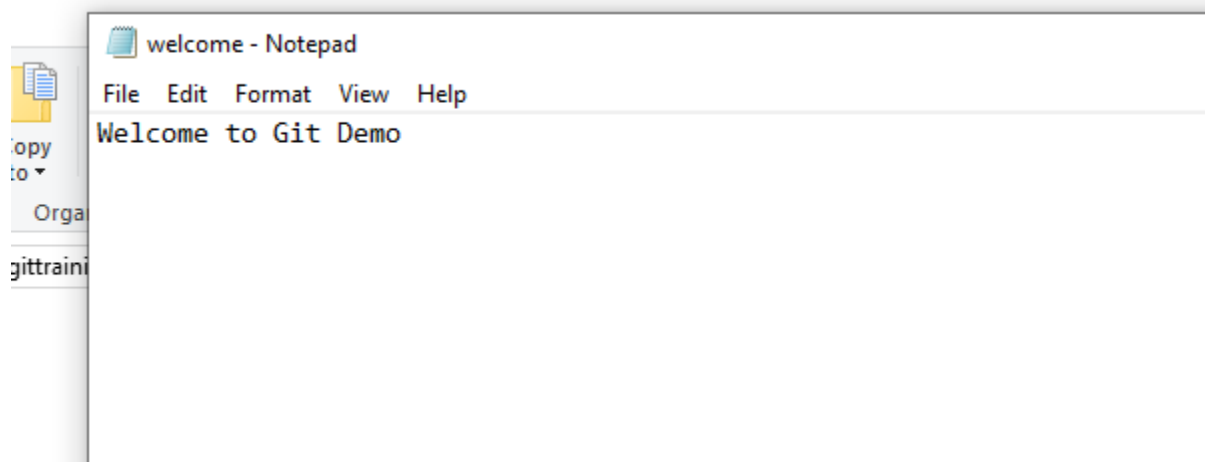
Output:

5 . To verify the content, execute the command cat filename

Solution:

```
narendrakumar.s@PTPLL614 MINGW64 /d/gittraining/GitDemo (main)
$ cat welcome.txt
Welcome to Git Demo
```

Output:



6. Execute git status

Solution:

```
narendrakumar.s@PTPLL614 MINGW64 /d/gittraining/GitDemo (main)
$ git add welcome.txt
warning: in the working copy of 'welcome.txt', LF will be replaced by CRLF the next time Git touches it

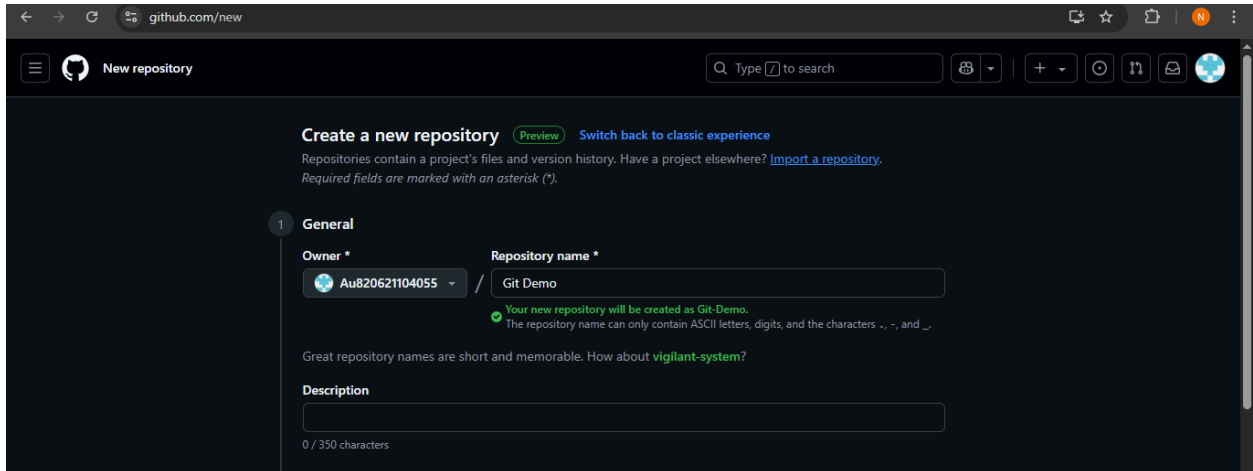
narendrakumar.s@PTPLL614 MINGW64 /d/gittraining/GitDemo (main)
$ git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   welcome.txt
```

7. Signup with GitHub and create a remote repository “**GitDemo**”

Solution:



8.Add remote origin

Solution:

```
narendrakumar.s@PTPLL614 MINGW64 /d/gittraining/GitDemo (main)
$ git remote add origin https://github.com/Au820621104055/GitDemo.git
```

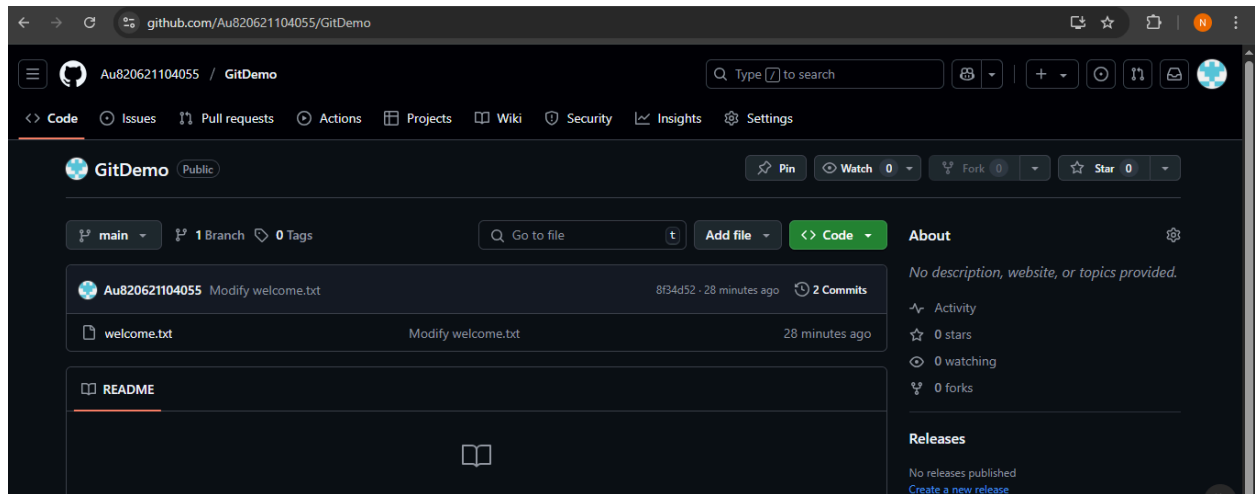
9. Create files and and push the local to remote repository.

Solution:

```
narendrakumar.s@PTPLL614 MINGW64 /d/gittraining/GitDemo (main)
$ git commit -m "Added welcome.txt"
[main (root-commit) 6a03f3f] Added welcome.txt
1 file changed, 1 insertion(+)
create mode 100644 welcome.txt

narendrakumar.s@PTPLL614 MINGW64 /d/gittraining/GitDemo (main)
$ git push origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 245 bytes | 122.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Au820621104055/GitDemo.git
* [new branch]      main -> main
```

Output:



10. Make changes in the remote and pull the changes to the local

Solution:

```
narendrakumar.s@PTPLL614 MINGW64 /d/gittraining/GitDemo (main)
$ git pull origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 949 bytes | 47.00 KiB/s, done.
From https://github.com/Au820621104055/GitDemo
* branch                main          -> FETCH_HEAD
   6a03f3f..8f34d52      main          -> origin/main
Updating 6a03f3f..8f34d52
Fast-forward
 welcome.txt | 1 +
 1 file changed, 1 insertion(+)
```

11. Create a .gitignore file to ignore contents of temp folder

Solution:

```

narendrakumar.s@PTPLL614 MINGW64 /d/gittraining/GitDemo (main)
$ mkdir temp

narendrakumar.s@PTPLL614 MINGW64 /d/gittraining/GitDemo (main)
$ echo "ignore this file" > temp/file1.txt

narendrakumar.s@PTPLL614 MINGW64 /d/gittraining/GitDemo (main)
$ echo "temp/" > .gitignore

narendrakumar.s@PTPLL614 MINGW64 /d/gittraining/GitDemo (main)
$ git add .gitignore
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it

narendrakumar.s@PTPLL614 MINGW64 /d/gittraining/GitDemo (main)
$ git commit -m "Added the .gitignore file "
[main e2c18de] Added the .gitignore file
2 files changed, 2 insertions(+)
create mode 100644 .gitignore
create mode 100644 temp.gitignore

```

Docker

1. What is Docker-compose.yml file? Give sample docker-compose file

Solution:

- docker-compose.yml is a configuration file used by Docker Compose to define and run multi-container Docker applications
- Instead of running multiple docker run commands manually, you define all services (containers), networks, and volumes in one file.

Example

```

version: "3.8"

services:
  web:
    image: nginx:latest
    ports:
      - "8080:80"

  db:
    image: mysql:5.7
    environment:
      MYSQL_ROOT_PASSWORD: root
      MYSQL_DATABASE: testdb
    volumes:
      - db_data:/var/lib/mysql

volumes:
  db_data:

```

2. What is Docker hub?
 - a. Docker Hub is a cloud-based registry where you can find, share, and store Docker images.
 - b. It has official images like nginx, mysql, ubuntu, etc.
 - c. we can also push your own custom images to Docker Hub.

Example:

```
docker pull nginx
docker login
docker tag myapp username/myapp:v1
docker push username/myapp:v1|
```